RS1 — The Vorton-Scalar and the hidden-axis singlet superfluid on the deflating 4+4 and 7+1 spacetimes

PROJECT spacetimes, Revision_Deflation. Supplement to both manuscripts of the revision: reduction to four dimensions, the deflation clock, modes of the scalar, pairing interaction, gap, bound pair, order parameter, vortices, and whether the paired state can follow the deflation

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November 12, 2026

0. Purpose, and how to read this supplement

The main project invented a complex scalar field, the **Vorton-Scalar**, whose exchange between the fermions of a 16-component Dirac field binds them into pairs, and showed that the pairs condense into a superfluid whose order parameter points along one of the four extra axes $x0, x5, x6, x7$ (supplement S1 of the main project, in the folder `supplement` above this one). All of that was done on flat spacetime.

This supplement repeats every calculation of S1 on the two spacetimes of the revision, in which the coordinates $x5$, $x6$, $x7$ deflate exponentially. It is written for a reader who has S1 at hand and who has read the document `dirac matrices deflating` (DMD), where the geometry and the Dirac equations of the deflating spacetimes are derived. The single most important fact from DMD is this (DMD, Sec. 3.3):

> *A field that does not depend on $x5, x6, x7$ obeys, after multiplication by the square root of the volume factor V_d, exactly the equations of flat spacetime.*

The superfluid of the main project lives in the sector of such fields. Its equations therefore do not change. What changes is the value of every coupling of the four-dimensional description, because a four-dimensional coupling is an eight-dimensional coupling divided by a power of the volume of the extra coordinates, and that volume now shrinks. The consequence is a *history*: quantities that were constants of the main project become functions of the ordinary time $x4$, and the thresholds of the main project — the coupling at which two fermions bind, at which matter becomes self-bound, at which the vacuum becomes unstable — are passed one after another, at definite times.

The sections follow those of S1. Section 1 reduces the theory to four dimensions and introduces the deflation clock. Section 2 treats the modes of the Vorton-Scalar itself. Section 3 gives the pairing interaction and the gap, Section 4 the bound pair, Section 5 the order parameter, Section 6 the vortices. Section 7 asks the question that has no counterpart on flat spacetime: *can the paired state follow the deflation?* Section 8 separates what has been established from what has not.

0.1 Symbols

The symbols of the main project keep their meanings (provenance P1 of the main project). The revision adds the following; the complete registry of the revision, with the proof that no symbol has two meanings, is the provenance RP1.

symbol	in programs and data files	meaning	defined in	origin
τ_d	`tau_d`	the deflation clock, $\tau_d = \mathrm{gsum}\,(x4 - x4_{\mathrm{ref}})$: the number of e-foldings by which the volume V_d has shrunk since the reference time (the letter τ without the index d is the optical depth of the main registry)	RS1 1; M V	new
τ_{pair}	`tau_pair`	the clock measured from the moment at which two fermions become bound: $\ln(D_{VS}/D_c)$	RS1 4; RS4 1; M VII	new
$\hat{\Phi}_{VS}$	`Phihat`	the weighted Vorton-Scalar $\sqrt{V_d}\,\Phi_{VS}$	RS1 1; M V	new

symbol	in programs and data files	meaning	defined in	origin
Ξ_d	`Xi_d`	$\mathrm{gsum}/(\lambda\,\Delta_{\mathrm{eq}})$: decides whether the paired state can follow the deflation ($\Xi_d \ll 1$: yes)	RS1 7; M VIII	new
n_k	—	occupation of the pair of states of the single-particle energy ξ_k in the shell model of the paired state (with the index k; the letter n without index is a density)	RS1 7; M VIII	new
Δ_{eq}	`gap_eq`	the equilibrium gap for the pairing strength of the moment	RS1 7; M VIII	new
E_{sh}	`E_sh`	half-width in energy of the shell of states around the Fermi surface in the model of the pair dynamics; unit of energy and of inverse time there	RS1 7; M VIII	new
n_{ex}	`n_ex`, `n_ex_final`	density of broken pairs (in units of E_{sh} times the density of states)	RS1 7; M VIII	new

symbol	in programs and data files	meaning	defined in	origin
P_c	`Pc`	$i\alpha^1\alpha^2\alpha^3\chi$, the product of the five matrices of the sector that does not depend on the extra coordinates; equal, up to sign, to the generator P of $U(1)_E$ of the main registry	RS1 6; M X	new
θ_{12}	—	the angle around a vortex line that lies along $x3$, in the plane of $x1$ and $x2$ (with the index 12; θ with other indices and the scattering angle keep the meanings of the main project)	RS1 6; M X	new
`M` (main registry)	`M_gap`	the 8×8 gap matrix of the main registry; `M_gap` is its spelling in the statements of the programs of the revision	RS1 5; M X	main
w_z	`w_z`	a constant spinor of a fermion zero mode on a vortex line	RS1 6; M X	new

Symbols of DMD that are used here: the rates g55, g66, g77 and their sum gsum; the scale factors b_5, b_6, b_7 and the volume factor $V_d = b_5 b_6 b_7 \propto e^{-\mathrm{gsum}\,x4}$; the weighted field $\hat{\psi} = \sqrt{V_d}\,\psi$; the wavenumbers k_t along the deflating coordinates, the physical wavenumber k_d and the variable $u_d = k_d/\mathrm{rate}_t$; the temperature $T_{\mathrm{defl}} = \mathrm{rate}_t/2\pi$.

0.2 How to reproduce the numbers

```
cd "/Users/nsh/Developer/github/spacetimes/Revision_Deflation"
cargo build --release --manifest-path code/deflation-rs/Cargo.toml
./code/deflation-rs/target/release/deflation symmetry    # reduction of the couplings, vortices
```

```
./code/deflation-rs/target/release/deflation channels    # exchange amplitudes, modes of the
↪  scalar
./code/deflation-rs/target/release/deflation history     # gap, bound pair, order of events
./code/deflation-rs/target/release/deflation quench      # dynamics of the paired state
python3 code/scripts/make_figures.py clock scalar kernels gap binding quench
python3 code/scripts/write_doc_tables.py                 # the tables of this supplement
python3 code/scripts/assemble_docs.py docs_src/RS1_vorton_scalar_and_superfluid_deflating.src.md
bash code/scripts/build_docs.sh supplement/RS1_vorton_scalar_and_superfluid_deflating.md
```

The command `history` takes about a minute and a half, the command `quench` a quarter of a minute, the others a few seconds. The complete output of the four commands is reproduced in Appendix A.

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1. Reduction to four dimensions, and the deflation clock

1.1 The Lagrangian on the deflating spacetimes

On a curved spacetime a Lagrangian is multiplied by $\sqrt{|\det g|} = V_d$, partial derivatives of spinors become covariant derivatives, and the flat metric is replaced by the curved one. For the 7+1 model of S1, Eq. (1.1), this gives

$$\begin{aligned}\mathcal{L}^{VS}_{1671d} = V_d \Big[&\bar\psi\, i\gamma^\mu \nabla_\mu \psi + g^{\mu\nu}\, \partial_\mu \Phi^*_{VS}\, \partial_\nu \Phi_{VS} - \lambda_{VS}\Big(|\Phi_{VS}|^2 - \frac{v_{VS}^2}{2}\Big)^2 \\ &- y_{VS}(\Phi_{VS}\, \bar\psi P^R \psi + \Phi^*_{VS}\, \bar\psi P^L \psi)\Big],\end{aligned} \tag{1.1}$$

with $\psi = \psi_{1671d}$, the matrices $\gamma^\mu = \gamma^\mu_{1671d}$ and the metric $g = g_{1671d}$ of DMD. In words: Φ_{VS} is the Vorton-Scalar, a complex scalar field; λ_{VS} is its self-coupling and $v_{VS}/\sqrt{2}$ the magnitude at which its potential is lowest; y_{VS} is its Yukawa coupling to the fermions; and $P^{L,R}$ are the chiral projectors. The three constants λ_{VS}, v_{VS}, y_{VS} are constants of the eight-dimensional theory and *do not change*.

The symmetries $U(1)_F$ (fermion number) and $U(1)_{VS}$ (the chiral phase symmetry that the vacuum value of the Vorton-Scalar breaks) of S1 are symmetries of Eq. (1.1) as well, because the transformations do not involve the coordinates.

The vacuum. A constant field $\Phi_{VS} = v_{VS}/\sqrt{2}$ with $\psi = 0$ solves the field equations of Eq. (1.1) on the deflating spacetime, because a constant has no derivatives. The vacuum value of the eight-dimensional field is the same at all times.

1.2 Fields that do not depend on the extra coordinates

As in Stage 1 of the main project, consider fields that do not depend on the four extra coordinates $x0, x5, x6, x7$. Introduce the weighted fields

$$\hat\psi = \sqrt{V_d}\,\psi, \qquad \hat\Phi_{VS} = \sqrt{V_d}\,\Phi_{VS}.$$

Each term of Eq. (1.1) is now rewritten.

- *Fermion kinetic term.* By DMD, Eq. (3.6), $V_d\,\bar\psi\, i\gamma^\mu\nabla_\mu\psi = \bar{\hat\psi}\, i\gamma^A\partial_A\hat\psi$ with the constant matrices of the main project: exactly the flat term.
- *Yukawa term.* It contains one scalar and two fermion fields, hence $V_d \cdot V_d^{-3/2} = V_d^{-1/2}$: $V_d\, y_{VS}\, \Phi_{VS}\bar\psi P\psi = (y_{VS}/\sqrt{V_d})\, \hat\Phi_{VS}\bar{\hat\psi}P\hat\psi$.
- *Potential.* $V_d\,\lambda_{VS}(|\Phi_{VS}|^2 - v_{VS}^2/2)^2 = (\lambda_{VS}/V_d)\,(|\hat\Phi_{VS}|^2 - V_d\, v_{VS}^2/2)^2$.

- *Scalar kinetic term.* The derivatives along $x1, x2, x3$ give the flat term for $\hat{\Phi}_{VS}$. The derivative along the time acts on the weight as well: $\sqrt{V_d}\,\partial_4\Phi_{VS} = \partial_4\hat{\Phi}_{VS} + \frac{1}{2}\mathrm{gsum}\,\hat{\Phi}_{VS}$, so that

$$V_d\,|\partial_4\Phi_{VS}|^2 = |\partial_4\hat{\Phi}_{VS}|^2 + \frac{\mathrm{gsum}^2}{4}\,|\hat{\Phi}_{VS}|^2 + \frac{\mathrm{gsum}}{2}\,\partial_4|\hat{\Phi}_{VS}|^2.$$

  The last term is a total derivative and does not affect the equations of motion.

The result is the Lagrangian of S1 for the weighted fields, with three changes (check `def-S4`, exact arithmetic):

$$\boxed{\begin{aligned} &y_{VS} \to \frac{y_{VS}}{\sqrt{V_d(x4)}}, \qquad \lambda_{VS} \to \frac{\lambda_{VS}}{V_d(x4)}, \qquad v_{VS} \to v_{VS}\sqrt{V_d(x4)}, \\ &\text{and the additional term } + \frac{\mathrm{gsum}^2}{4}\,|\hat{\Phi}_{VS}|^2. \end{aligned}} \tag{1.2}$$

The first three are the four-dimensional couplings of S1, Sec. 1.3, where the volume of the extra coordinates was a constant V_4; here that volume is the coordinate volume of the cell times $V_d(x4)$, and it shrinks. The fourth term is new. It lowers the square of the mass of each mode of the scalar by $\mathrm{gsum}^2/4$ in the equation of the weighted field; Sec. 2 shows what it does and what it does not do.

The 4+4 model. The main project defined the Vorton-Scalar of the 4+4 spacetime only in the sector of fields that do not depend on the additional times $x5, x6, x7$ (S1, Eq. (3.1)), and stated plainly that on flat spacetime this restriction is imposed, not derived. On the deflating 4+4 spacetime the restriction has a dynamical reason: every admissible mode that depends on a deflating time is extinguished (DMD, Sec. 4.3). In that sector the weighted fields obey the equations of S1 with the same replacements (1.2); the Lagrangian $\mathcal{L}^{VS}_{1644d}$ is Eq. (3.1) of S1 written for $\hat{\psi}$ and $\hat{\Phi}_{VS}$ with the couplings (1.2).

1.3 What changes, and what does not

Define the **deflation clock**

$$\tau_d = \mathrm{gsum}\,(x4 - x4_{\mathrm{ref}}), \tag{1.3}$$

the number of e-foldings by which the volume V_d has shrunk since a reference time $x4_{\mathrm{ref}}$, which is chosen in each application and stated there. By Eq. (1.2) every quantity of the four-dimensional description is proportional to $e^{n\tau_d}$ with a definite exponent n:

quantity	meaning	exponent n
V_d	$b_5b_6b_7$: the factor $\sqrt{\lvert\det g\rvert}$ of the measure	-1
y_{VS} of the four-dimensional description	Yukawa coupling of the weighted fields	$\frac{1}{2}$
λ_{VS} of the four-dimensional description	quartic coupling of the weighted Vorton-Scalar	1
v_{VS} of the four-dimensional description	vacuum value of the weighted Vorton-Scalar	$-\frac{1}{2}$
m	fermion mass $y_{VS}v_{VS}/\sqrt{2}$	0
m_s	mass of the radial mode $\sqrt{2\lambda_{VS}}\,v_{VS}$	0
g_{VS}	coupling of the radial mode to the fermions, m/v_{VS}	$\frac{1}{2}$
α_{VS}	$g^2_{VS}/4\pi$	1
G_{VS}	constant of the Fermi type of the neutral-current interaction, $1/(\sqrt{2}\,v^2_{VS})$	1

quantity	meaning	exponent n
D_{VS}	$\alpha_{VS}\, m/m_s$: decides whether two fermions bind	1
C	$g_{VS}^2 m^2/m_s^2$: decides the state of matter	1
vortex energy per length	proportional to v_{VS}^2 (four-dimensional)	-1
G_{Fermi} (Fermi constant of the weak interaction) if the vacuum value of the Vorton-Scalar sets the electroweak scale (assumption HV)	$1/(\sqrt{2}\, v_{VS}^2)$, four-dimensional vacuum value	1
G_N (Newton's constant) if gravity is eight-dimensional (A2)	G_8 divided by the volume of $x5, x6, x7$	1

The last two rows hold under additional hypotheses, which the supplements RS2 and RS3 state and examine. Three observations organize everything that follows.

1. **The masses do not change.** The fermion mass $m = y_{VS} v_{VS}/\sqrt{2}$ and the mass $m_s = \sqrt{2\lambda_{VS}}\, v_{VS}$ of the radial mode contain the couplings in combinations from which the volume cancels. So do the Fermi momentum and the Fermi energy of matter with a given number of particles per unit of ordinary volume.
2. **The coupling grows.** The coupling $g_{VS} = m/v_{VS}$ of both modes of the scalar to the fermions grows as $e^{\tau_d/2}$, because the four-dimensional vacuum value falls. The strength of the attraction, $\alpha_{VS} = g_{VS}^2/4\pi$, grows as e^{τ_d}: it doubles every time the volume of the deflating coordinates halves.
3. **The two numbers that decide the state of matter grow at the same rate.** These are $D_{VS} = \alpha_{VS}\, m/m_s$, which decides whether two fermions form a bound pair, and $C = g_{VS}^2 m^2/m_s^2$, which decides whether matter keeps its fermion mass. Their ratio $C/D_{VS} = 4\pi\, m/m_s$ is fixed.

A model of the main project is specified by the masses and one coupling. On a deflating spacetime it is specified by the masses and the value of the coupling *at one time*; the deflation then carries it through all larger couplings. Panel (a) of the figure shows the exponents; panel (b) shows the order in which the thresholds of the main project are passed, which Sec. 4 and the supplement RS4 derive.

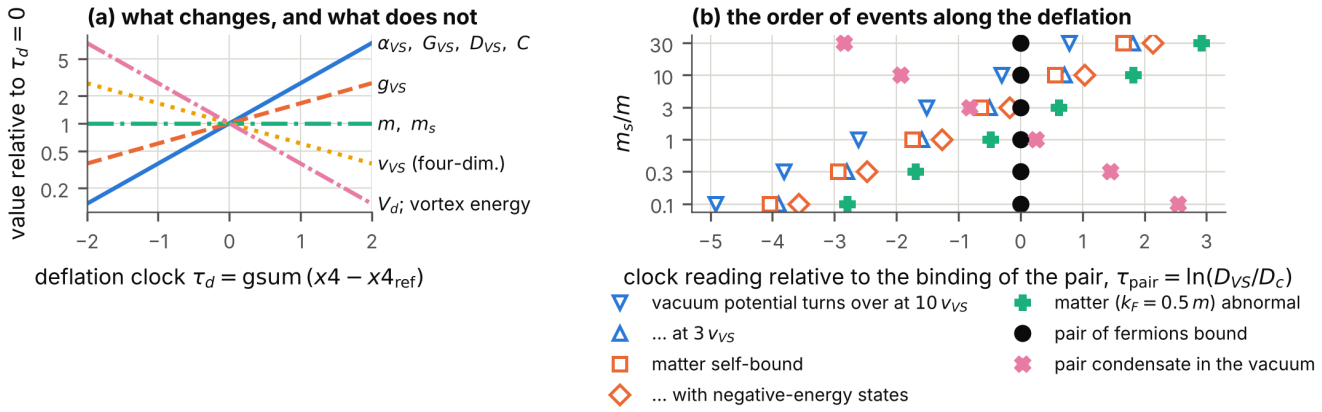


Figure 1: The deflation clock. (a) How the quantities of the four-dimensional description change with the clock τ_d (logarithmic scale; values relative to $\tau_d = 0$). (b) The order of events along the deflation for six ratios m_s/m: clock reading of each event relative to the moment at which two fermions form a bound pair. Data: `data/coupling_exponents.csv`, `data/events.csv`; programs: `deflation symmetry`, `deflation history`.

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2. The modes of the Vorton-Scalar on the deflating spacetimes

Write the Vorton-Scalar as in S1, Eq. (1.3), in terms of its **radial mode** (a real field of mass m_s) and its **phase mode** (a real massless field, the Goldstone boson of the broken $U(1)_{VS}$). Let $\varphi_{\rm bos}$ stand for either of them, with the mass $m_{\rm bos} = m_s$ or $m_{\rm bos} = 0$, and let $\hat{\varphi}_{\rm bos} = \sqrt{V_d}\,\varphi_{\rm bos}$ be its weighted form. To first order in the field, its equation on the deflating spacetime is

$$\frac{1}{V_d}\,\partial_4(V_d\,\partial_4\varphi_{\rm bos}) + \Big(k^2 + \eta_{44}\sum_t \frac{k_t^2}{b_t^2} + m_{\rm bos}^2\Big)\varphi_{\rm bos} = 0, \tag{2.1}$$

for a mode with the wavenumber of magnitude k along the flat coordinates and the wavenumbers k_t along the deflating ones. The sign η_{44} is $+1$ on 7+1 and -1 on 4+4: on 4+4 the deflating coordinates are time-like and their wavenumbers enter with the opposite sign, as for the fermions.

2.1 Modes that do not depend on the deflating coordinates

With $k_t = 0$, Eq. (2.1) reads $\varphi''_{\rm bos} - {\rm gsum}\,\varphi'_{\rm bos} + (k^2 + m_{\rm bos}^2)\varphi_{\rm bos} = 0$, a linear equation with constant coefficients. Its solutions are exponentials,

$$\varphi_{\rm bos} \;\propto\; \exp\Big\{\Big[\frac{\rm gsum}{2} \pm \sqrt{\frac{{\rm gsum}^2}{4} - k^2 - m_{\rm bos}^2}\,\Big]\,x4\Big\} \tag{2.2}$$

(check `def-B2`). Define

$$\varpi_b = \sqrt{k^2 + m_{\rm bos}^2 - \frac{{\rm gsum}^2}{4}}.$$

- If $k^2 + m_{\rm bos}^2 > {\rm gsum}^2/4$, the mode **oscillates** with the frequency ϖ_b, and its amplitude grows as $e^{{\rm gsum}\,x4/2}$. The growth is that of the weight alone: $\hat{\varphi}_{\rm bos}$ has a constant amplitude. The number of quanta in a cell of fixed coordinates is conserved, and the cell shrinks.
- If $k^2 + m_{\rm bos}^2 < {\rm gsum}^2/4$, the mode **does not oscillate**. For the radial mode this never happens if $m_s > {\rm gsum}/2$, which holds by many orders of magnitude for any rate of deflation that is not excluded by observation (RS3). For the phase mode it happens for $k < {\rm gsum}/2$, that is, for wavelengths longer than $4\pi/{\rm gsum}$.

What the non-oscillating phase modes mean. At $k = 0$ the two exponents are 0 and gsum. The first is a constant phase, which costs nothing. The second is a phase that advances at a rate proportional to $1/V_d$: it is the conservation of the charge of $U(1)_{VS}$ in a shrinking volume. Neither is an instability of the vacuum.

2.2 The field of a static source: the potential between two fermions

Two fermions at rest attract each other by exchanging the radial mode. On the deflating spacetime the density of a fixed number of fermions grows as $1/V_d$. The radial mode therefore obeys Eq. (2.1) with a source proportional to $e^{{\rm gsum}\,x4}$, and the solution that follows the source is

$$\varphi_{\rm bos} = \frac{e^{{\rm gsum}\,x4}}{k^2 + m_{\rm bos}^2} \times (\text{source}),$$

as substitution shows: the terms ${\rm gsum}^2 - {\rm gsum}\cdot{\rm gsum}$ cancel. The denominator contains the *unshifted* mass. The potential between two fermions at rest is therefore, at every moment, the

Yukawa potential of S1,

$$V_s(r) = -\alpha_{VS}(x4)\,\frac{e^{-m_s r}}{r}, \tag{2.3}$$

with the range $1/m_s$ unchanged and the strength of the moment. The numerical solution of Eq. (2.1) from rest approaches this form with a transient that decays as $e^{-\tau_d/2}$; after $\tau_d = 20$ the relative difference is below 4×10^{-5} in the four cases computed (check `def-B1`).

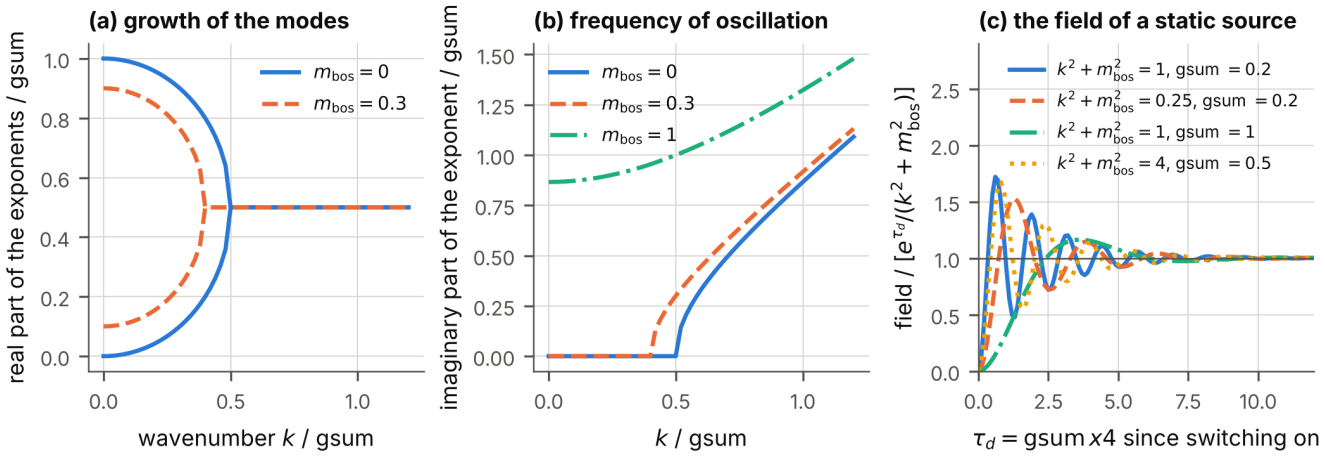


Figure 2: Modes of the Vorton-Scalar that do not depend on the deflating coordinates. (a) Real parts of the two exponents of Eq. (2.2) against the wavenumber, for the phase mode and for a radial mode, in units of gsum. (b) Their imaginary part, the frequency of oscillation. (c) The field that a static source builds up from zero, divided by the form $e^{\tau_d}/(k^2 + m_{\rm bos}^2)$ that leads to Eq. (2.3). Data: `data/scalar_zero_modes.csv`, `data/static_source.csv`; program: `deflation channels`.

2.3 Modes that depend on a deflating coordinate

For $k_t \neq 0$ the weighted field obeys $\hat{\varphi}''_{\rm bos} + (\varpi_b^2 + \eta_{44}\, k_d(x4)^2)\hat{\varphi}_{\rm bos} = 0$, with the physical wavenumber k_d that grows exponentially. The two spacetimes behave as they do for the fermions (DMD, Part 4).

7+1: creation of quanta. The frequency grows without bound. When one rate governs, the solutions are Bessel functions of the imaginary order $i\varpi_b/\text{rate}_t$ and of the argument u_d, and a mode that starts empty ends with

$$n_{\rm cr} = \frac{1}{e^{2\pi\varpi_b/\text{rate}_t} - 1} \tag{2.4}$$

quanta: the Bose–Einstein distribution at the temperature $T_{\rm defl} = \text{rate}_t/2\pi$, the same temperature as for the fermions. The program confirms Eq. (2.4) in 36 cases to a relative deviation of 1.9×10^{-5} (check `def-B3`; independently `Idef-B1`). The figure of DMD, Sec. 4.2, shows both laws.

4+4: extinction. Beyond the turning value at which $k_d = \varpi_b$ one solution decays and one grows. The bounded solution is the modified Bessel function of the order $i\varpi_b/\text{rate}_t$ that decays at large argument. At early times it is a standing wave; at late times it falls as e^{-u_d} (check `def-B4`). The scalar field, like the fermion field, loses its dependence on the deflating times.

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3. The pairing interaction and the gap along the clock

3.1 Exchange amplitudes

S1, Sec. 4, defines for every pair channel the exchange amplitudes F_S and F_P of the radial mode and of the phase mode: the matrix elements of the one-boson exchange between two fermions of

opposite momenta in the internal state of the channel. They are computed from the Dirac matrices of the frame, which the deflation does not change, and each is proportional to the square of the coupling. Therefore, for fields that do not depend on $x5, x6, x7$,

$$F_{S,P} \text{ on the deflating spacetime at the time } x4 \;=\; g_{VS}(x4)^2 \times (F_{S,P} \text{ of S1 at } g_{VS} = 1). \qquad (3.1)$$

The program computes the amplitudes of the representative pair channels of the main project again — 15 on each spacetime, at three momenta and 13 scattering angles — from the matrices of each deflating spacetime, and compares them with the file of the main project: the 1170 rows agree to the digits printed there (check `def-K1`), and the amplitudes of the hidden-axis channel reproduce the closed forms of S1 to 1.1×10^{-15} (check `def-K2`).

The conclusion of S1 about the channels therefore stands at every moment: the radial mode attracts in every channel; the phase mode attracts in the **vector pairs along the extra axes** and repels in their two-form partners; the most attractive channel is the vector pair along an extra axis, the *hidden-axis singlet*.

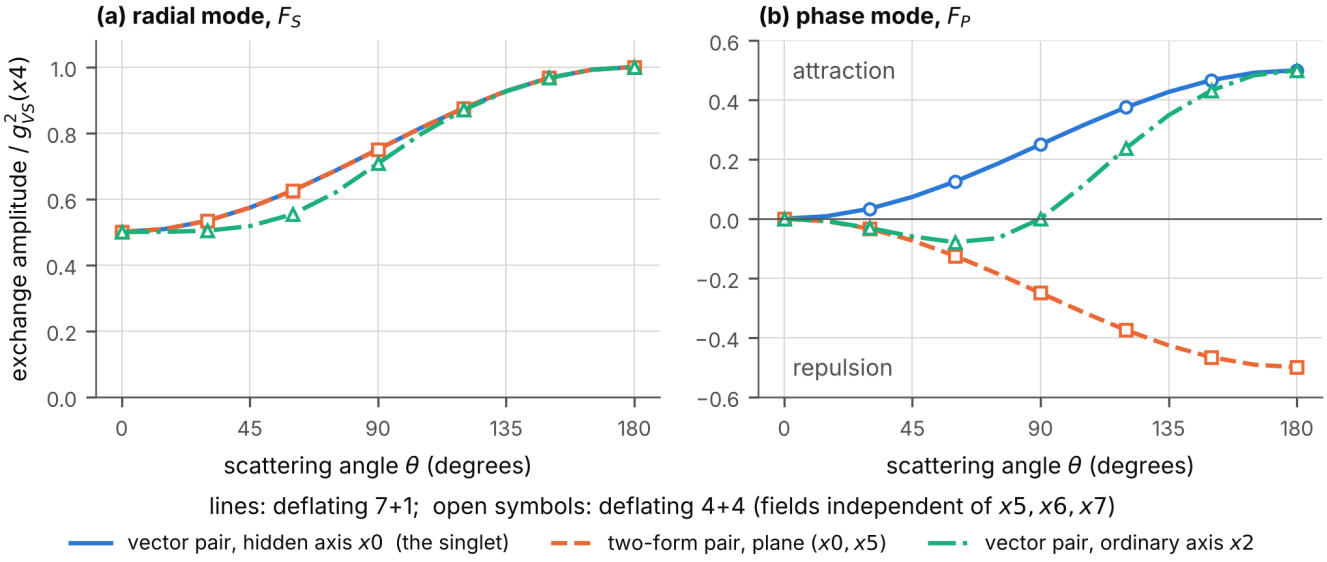


Figure 3: Exchange amplitudes in three pair channels, divided by the square of the coupling of the moment, against the scattering angle. Lines: 7+1; open symbols: 4+4. The amplitudes are those of the main project: the deflation changes their overall strength and nothing else. Data: `data/pair_channel_kernel.csv`; program: `deflation channels`.

3.2 The gap grows, and much faster than the coupling

The gap equation of S1, Sec. 5, holds at each moment with the couplings of that moment. Its solution as a function of the clock is the **adiabatic history** of the paired state: the state that matter of a given density would have if it were always in equilibrium. (Section 7 examines when it can be.) The program solves the gap equation at 34 moments for each of four sets of the mass m_s and the Fermi momentum k_F. Recall the dimensionless pairing strength $\lambda = N(0)\,W(k_F, k_F)$ of S1 (density of states at the Fermi surface times the kernel of the gap equation there); in the vector channel it is the sum $\lambda_S + \lambda_P$ of the contributions of the two modes.

m_s/m, k_F/m	τ_d	g_{VS}	λ	gap$/E_F$	T_c/gap	rate of growth	$\lambda \times$ rate
0.10, 0.50	-0.63	0.730	0.062	1.17×10^{-7}	0.567	—	—
0.10, 0.50	-0.13	0.939	0.102	7.40×10^{-5}	0.567	9.88	1.01

m_s/m, k_F/m	τ_d	g_{VS}	λ	gap$/E_F$	T_c/gap	rate of growth	$\lambda \times$ rate
0.10, 0.50	0.38	1.208	0.169	3.70×10^{-3}	0.567	6.01	1.01
0.10, 0.50	0.88	1.554	0.279	0.0407	0.565	3.74	—
0.10, 0.50	1.39	2.000	0.463	0.189	0.555	2.50	—
0.10, 0.50	1.89	2.573	0.766	0.563	0.528	1.96	—
0.10, 0.50	2.39	3.311	1.268	1.49	0.480	1.96	—
1.0, 0.50	0.88	1.554	0.056	1.02×10^{-7}	0.567	—	—
1.0, 0.50	1.13	1.763	0.072	5.62×10^{-6}	0.567	14.02	1.01
1.0, 0.50	1.39	2.000	0.092	1.27×10^{-4}	0.567	10.93	1.01
1.0, 0.50	1.64	2.269	0.119	1.45×10^{-3}	0.567	8.54	1.02
1.0, 0.50	1.89	2.573	0.153	9.76×10^{-3}	0.567	6.71	1.03
1.0, 0.50	2.14	2.919	0.197	0.0439	0.566	5.33	—
1.0, 0.50	2.39	3.311	0.253	0.147	0.563	4.37	—

(The rate of growth in the last two columns is $d\ln(\text{gap})/d\tau_d$; the last column is given where the gap is between 10^{-6} and 0.02 of E_F.) Three results (checks `def-H2`, `def-H3`):

1. *The order of the channels does not change.* At every moment the gap of the vector channel is at least as large as the gap with the radial mode alone, and that at least as large as the gap of the two-form channel. The gap never decreases.
2. *The critical temperature follows the gap.* Wherever the gap is below 5 per cent of the Fermi kinetic energy E_F, the critical temperature is $T_c = 0.567$ times the gap, the weak-coupling ratio, to within 0.002.
3. *The law of growth.* In weak coupling the gap is proportional to $e^{-1/\lambda}$, and $\lambda \propto e^{\tau_d}$. Hence

$$\frac{d\ln(\text{gap})}{d\tau_d} = \frac{1}{\lambda}. \tag{3.2}$$

   In the computed histories the product $\lambda\, d\ln(\text{gap})/d\tau_d$ lies between 1.00 and 1.03 wherever the gap is between 10^{-6} and 0.02 of E_F.

Equation (3.2) is the central quantitative statement of this supplement. While the coupling grows by one e-folding, the gap of a weakly paired state with $\lambda = 0.1$ grows by ten e-foldings, a factor of twenty thousand. A paired state that is unobservably weak at one time becomes strong within a few e-foldings of the volume.

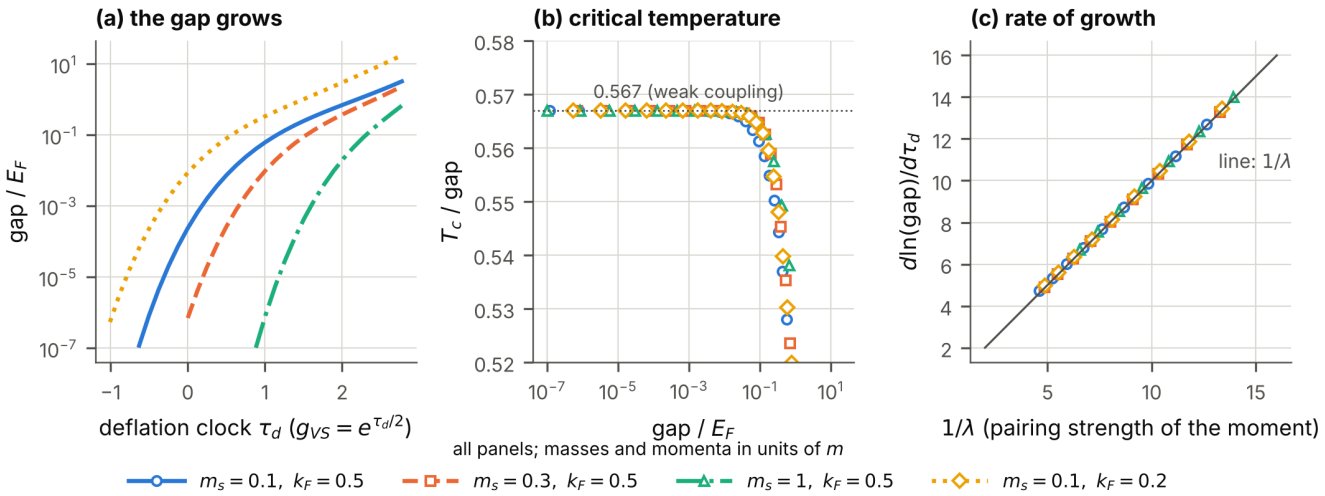


Figure 4: The gap along the deflation clock, for four sets of the mass of the radial mode and of the Fermi momentum (units of m). (a) The gap of the vector channel in units of the Fermi kinetic energy. (b) The ratio of the critical temperature to the gap. (c) The rate of growth of the gap against $1/\lambda$; the line is Eq. (3.2). Data: `data/gap_history.csv`; program: `deflation history`.

---

4. The bound pair

4.1 The moment of binding

Two fermions in the Yukawa potential (2.3) form a bound state if and only if $D_{VS} = \alpha_{VS}\, m/m_s$ exceeds the critical value $D_c = 1.67981$ (S1, Sec. 5.3; check `def-H1`). On a deflating spacetime $D_{VS} \propto e^{\tau_d}$, so the pair becomes bound at a definite time. Measure the clock from that moment:

$$\tau_{\text{pair}} = \ln \frac{D_{VS}}{D_c} = \text{gsum}\,(x4 - x4_{\text{binding}}). \tag{4.1}$$

τ_{pair}	D_{VS}	scattering length $\times\, m_s$	bound states	$E_b \times m/m_s^2$
-0.105	1.5118	-12.50	0	0
-0.095	1.5279	-13.93	0	0
-0.010	1.6630	-134.0	0	0
0.010	1.6966	135.9	1	5.50×10^{-5}
0.486	2.7314	2.698	1	0.232
0.994	4.5409	-0.1288	1	1.82

The scattering length. Just before the binding the scattering length a_{sc} of two fermions is large and negative, just after it large and positive: $-134.0/m_s$ at $0.99\,D_c$ and $+135.9/m_s$ at $1.01\,D_c$ (check `def-H4`). At the moment of binding it is infinite: the two fermions are at a scattering resonance. Every pair of fermions of the model passes through this resonance once, at the same time. The supplement RS3 examines what that means if the fermions are the dark matter.

The binding energy. Just after the binding the binding energy starts quadratically,

$$E_b \;\propto\; \left(e^{\tau_{\text{pair}}} - 1\right)^2 \frac{m_s^2}{m}, \tag{4.2}$$

with the fitted exponents 2.01 between 2 and 4 per cent and between 4 and 8 per cent above D_c. Because E_b vanishes at the threshold, the time $1/E_b$ that the pair needs to adjust is infinite there. *However slow the deflation is, the pair cannot follow it adiabatically at first.*

From the gap equation. The same threshold appears in the gap equation of the paired state in vacuum, whose kernel is relativistic: its linearized form reaches the eigenvalue 1 at $D_{VS} = 1.7347$, 1.6902, 1.6810 for $m_s/m = 0.3$, 0.1, 0.03, approaching 1.67981 as the binding becomes non-relativistic (check `def-P6` of the supplement RS4).

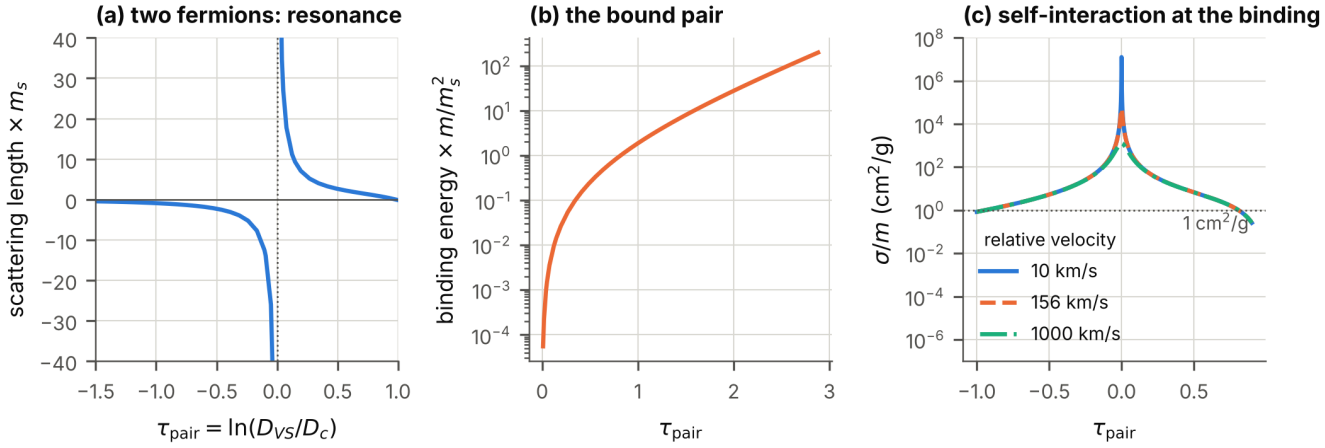


Figure 5: The bound pair along the clock. (a) Scattering length of two fermions in units of $1/m_s$ against τ_{pair}: the resonance at the moment of binding. (b) Binding energy of the pair. (c) Cross-section per mass of two fermions at three relative velocities while the clock passes the binding (supplement RS3). Data: `data/dimer_history.csv`, `data/dm_resonance.csv`; programs: `deflation history`, `deflation matter`.

4.2 The order of events

The other thresholds of the main project are thresholds in $C = g_{VS}^2 m^2/m_s^2$, which grows at the same rate as D_{VS}. With $C/D_{VS} = 4\pi m/m_s$ fixed, the order in which the events happen depends on the ratio m_s/m alone. The table gives the clock reading τ_{pair} of each event; a negative number means that the event precedes the binding of the pair.

event	threshold	$m_s/m =$ 0.1	$m_s/m =$ 0.3	$m_s/m =$ 1	$m_s/m =$ 3	$m_s/m =$ 10	$m_s/m =$ 30
the one-loop potential of the Vorton-Scalar falls below its vacuum value at the field $10\,v_{VS}$	$C = 1.5477$	-4.92	-3.82	-2.61	-1.51	-0.31	0.79
matter is self-bound (Fermi sea only)	$C = 3.7011$	-4.04	-2.94	-1.74	-0.64	0.56	1.66
the one-loop potential falls below its vacuum value at the field $3\,v_{VS}$	$C = 4.2694$	-3.90	-2.80	-1.60	-0.50	0.70	1.80
matter is self-bound (with the negative-energy states)	$C = 5.9218$	-3.57	-2.48	-1.27	-0.17	1.03	2.13
matter of Fermi momentum $0.5\,m$ becomes abnormal (massless fermions)	$C = 12.9713$	-2.79	-1.69	-0.49	0.61	1.82	2.91
two fermions form a bound pair	$D_{VS} = 1.6798$	0.00	0.00	0.00	0.00	0.00	0.00

event	threshold	$m_s/m =$ 0.1	$m_s/m =$ 0.3	$m_s/m =$ 1	$m_s/m =$ 3	$m_s/m =$ 10	$m_s/m =$ 30
pair condensate in the vacuum (ladder equation, cutoff $100\,m$)	$\alpha_{VS} = 2.13$ to 2.93	2.54	1.44	0.25	-0.82	-1.93	-2.84

The events other than the binding are derived in the supplement RS4. The table shows (check `def-H8`):

- For a **light radial mode** ($m_s/m \leq 3$ in the table) the pair becomes bound only *after* the one-loop potential of the Vorton-Scalar has fallen below its vacuum value at a field of three times the vacuum value: the deflation endangers the vacuum of the model before it binds the pair. The condition for the opposite order is $4\pi D_c\, m/m_s < 4.27$, that is $m_s/m > 4.9$.
- For a **heavy radial mode** ($m_s/m = 10$, 30) the pair becomes bound first, and matter becomes self-bound and abnormal later.

Which of the two orders nature would have chosen is not decided by anything in the model; the table states what follows from each choice.

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5. The order parameter and the ground state

5.1 Frame components and coordinate components

The order parameter of S1 is the expectation value Δ^a of the vector pair operator along the extra axis $a \in \{0, 5, 6, 7\}$: four complex numbers. On the deflating spacetimes the index a is a *frame* index: Δ^a is the component along the unit vector e_a. The coordinate components along the deflating coordinates are $\Delta_t = b_t(x4)\,\Delta^t$ and shrink with the scale factors when the frame components are constant. All statements below are about frame components.

The quartic invariant. The energy of S1 depends on the order parameter through the 8×8 gap matrix `M` on the internal states of a pair. The identity

$$\tfrac{1}{8}\,\mathrm{Tr}\big[(\mathsf{M}\mathsf{M}^\dagger)^2\big] = 2\Big(\sum_a |\Delta^a|^2\Big)^2 - \Big|\sum_{a,b} \eta_{ab}\,\Delta^a\Delta^b\Big|^2 \tag{5.1}$$

holds on both deflating spacetimes with the frame metric η_{ab} of the four axes (check `def-K3`, to 2×10^{-14}). In coordinate components the same expression contains the metric g_{tt} of the moment.

5.2 The ground state: no direction is preferred

The potential of the paired state as a function of the order parameter is computed in the supplement RS4. Its result for the six states examined on each spacetime follows. Each state is taken at the magnitude of the order parameter at which its potential is stationary. The squares of its two fourfold gaps are given in units of Δ_F^2, where Δ_F is the gap of a unitary state, and the potential in units of $N(0)\Delta_F^2$:

spacetime	state	smaller gap^2	larger gap^2	potential	flat directions	positive directions
7+1	pair along the frame axis of $x0$ (hidden-axis singlet)	1.000	1.000	-2.0000	4	4
7+1	pair along the frame axis of $x5$	1.000	1.000	-2.0000	4	4
7+1	pair along the frame axis of $x7$	1.000	1.000	-2.0000	4	4

spacetime	state	smaller gap^2	larger gap^2	potential	flat directions	positive directions
7+1	unitary, rotated between $x0$ and $x5$	1.000	1.000	-2.0000	4	4
7+1	unitary, general direction and phase	1.000	1.000	-2.0000	4	4
7+1	half-paired: $\eta_{ab}\Delta^a\Delta^b = 0$	0.000	1.000	-1.0000	—	—
4+4	pair along the frame axis of $x0$ (hidden-axis singlet)	1.000	1.000	-2.0000	4	4
4+4	pair along the frame axis of $x5$	1.000	1.000	-2.0000	4	4
4+4	pair along the frame axis of $x7$	1.000	1.000	-2.0000	4	4
4+4	unitary, rotated between $x0$ and $x5$	1.000	1.000	-2.0000	4	4
4+4	unitary, general direction and phase	1.000	1.000	-2.0000	4	4
4+4	half-paired: $\eta_{ab}\Delta^a\Delta^b = 0$	0.000	1.000	-1.0000	—	—

Every **unitary** state — a state in which Δ^a is a real unit vector times a phase — has all eight gaps equal and the lowest potential, whichever direction the vector has. It has four flat directions (the Goldstone modes: one phase and three rotations of the direction) and four directions of positive curvature. The **half-paired** states, in which four of the eight gaps vanish, have half the condensation energy. The potential does not distinguish the deflating axes from $x0$ (check `def-P4`).

What this means. One might have expected the deflation to select the direction of the order parameter: the hidden axis $x0$, which does not deflate, against the three that do. In the sector of fields that do not depend on the extra coordinates it does not. The reason is the central observation of DMD: in this sector the weighted field obeys the flat equations, whose internal symmetry $SU(2)_A \times SU(2)_B$ rotates the four frame axes into one another (check `def-S3`). The symmetry of the ground-state manifold is larger than the symmetry of the spacetime. The supplement RS2 takes up the consequences.

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6. Vortices and their zero modes

6.1 The vortex line of the Vorton-Scalar

Because the vacuum value of the Vorton-Scalar breaks the phase symmetry $U(1)_{VS}$, the field has line-like defects (S1, Sec. 8): along a straight line, say the $x3$ axis, its magnitude vanishes, and around the line its phase winds by 2π times an integer. For unit winding, $\Phi_{VS} = (v_{VS}/\sqrt{2}) \times$ (a profile that rises from 0 on the line to 1 at a distance of order $1/m_s$) $\times\, e^{i\theta_{12}}$, with θ_{12} the angle around the line in the plane of $x1$ and $x2$.

Energy per unit length. It is proportional to the square of the four-dimensional vacuum value, and so falls as $e^{-\tau_d}$: in the four-dimensional description **vortex lines become lighter as the deflation proceeds**, while their core, of radius $1/m_s$, keeps its size.

6.2 Fermions on the line: four zero modes for unit winding

On the line the fermion mass $m = g_{VS} \times$ (magnitude of the field) vanishes, and fermions can be trapped there. With $\Phi P^R + \Phi^* P^L = |\Phi|\, e^{i\chi \arg \Phi}$ the Hamiltonian of a fermion in the field of the vortex is

$$\sum_{j=1}^{3} \alpha^j p_j + m\,(\text{profile})\,\beta\,(\cos\theta_{12} + i\chi\,\sin\theta_{12}),$$

with the matrices α^j, β and the chirality matrix χ of the spacetime at hand, and the momenta p_j along $x1, x2, x3$. A state of zero energy at zero momentum along the line has the form of a constant spinor w_z times a function that decays away from the line as $\exp(-m \int \text{profile})$. Inserting this form, the Hamiltonian annihilates the state for every angle θ_{12} if and only if

$$(i\alpha^1 + \beta)\, w_z = 0, \qquad (i\alpha^2 + i\beta\chi)\, w_z = 0. \tag{6.1}$$

The five matrices $\alpha^1, \alpha^2, \alpha^3, \beta, i\beta\chi$ are Hermitian, have the square one and anticommute with one another (checks `44d-V1`, `71d-V1`). The two conditions (6.1) are the statements that w_z is an eigenvector, with the eigenvalue -1, of the two commuting matrices $i\beta\alpha^1$ and $\alpha^2\beta\chi$ (checks `..-V2`). Each joint eigenspace of the two matrices has the dimension $16/4 = 4$ (checks `..-V3`):

> **A vortex line of unit winding carries four zero modes of the sixteen-component field** (checks `44d-V5`, `71d-V5`; independently `I44d-V1`, `I71d-V1`).

Their motion along the line. For a momentum p_3 along the line the energy is the eigenvalue of $\alpha^3 p_3$. On the four zero modes α^3 equals the matrix

$$P_c = i\,\alpha^1\alpha^2\alpha^3\chi,$$

the product of the five matrices, which commutes with each of them, is Hermitian and traceless and has the square one; it is, up to its sign, the generator P of the symmetry $U(1)_E$ of the main project (checks `..-V4`). Its trace on the four zero modes vanishes: **two zero modes move along the line in one direction at the speed of light (energy $+p_3$) and two in the other (energy $-p_3$)**, and the direction of motion is tied to the charge under $U(1)_E$. For winding -1 the two directions are exchanged (checks `..-V6`).

On the deflating spacetimes. The equations (6.1) contain only the matrices of the frame and the fermion mass, neither of which changes. The four zero modes exist at every moment of the deflation, on both spacetimes. Massless fermions that run along a vortex line carry a current; a closed loop of the line can be held open by that current, and such loops are what the literature on cosmic strings calls vortons (S1, Sec. 8). Because the energy per length of the line falls as $e^{-\tau_d}$ while the fermions on it stay massless, the balance between the tension of a loop and the current on it shifts in favour of the current as the deflation proceeds. This supplement establishes the ingredients of that statement; it does not compute the equilibrium radius of a loop.

Relation to the main project. The provenance P11 of the main project records that its own count of these zero modes awaits correction, in a stage that the author has paused. The count given here was computed afresh for the revision; because it involves only the frame matrices, it holds on the flat spacetimes as well.

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7. Can the paired state follow the deflation?

7.1 The question, and the parameter that decides it

Section 3 gave the gap that the paired state would have in equilibrium at each moment. But a state can change at most at the rate of its own energy scale, which for a paired state is the gap. By Eq. (3.2) the equilibrium gap Δ_{eq} changes at the rate gsum/λ. The paired state can follow only if this rate is small compared with the gap itself, that is, only if

$$\Xi_d = \frac{\text{gsum}}{\lambda\,\Delta_{\text{eq}}} \;\ll\; 1. \tag{7.1}$$

Early in the deflation the gap is exponentially small and the condition fails, however small the rate is. The question is what happens then, and when the state catches up.

7.2 The model

The question is answered with the standard reduced model of a superfluid. Pairs of states of opposite momenta, with the single-particle energy ξ_k measured from the Fermi energy, fill a shell $-E_{\mathrm{sh}} < \xi_k < E_{\mathrm{sh}}$ with a constant density of states, and all attract one another with the dimensionless strength λ. In the mean-field approximation each pair of states is described by its pair correlation κ (a complex number) and its occupation n_k, and the order parameter is $\Delta = \lambda \int \kappa \, d\xi_k$. The equations of motion are

$$\frac{d\kappa}{dx4} = -i\left[2\,\xi_k\,\kappa - \Delta\,(1-2n_k)\right], \qquad \frac{dn_k}{dx4} = 2\,\mathrm{Im}\big(\Delta^{*}\kappa\big). \tag{7.2}$$

They conserve $|2\kappa|^2 + (1-2n_k)^2 = 1$ for every pair of states (check `def-Q1`). In equilibrium $\kappa = \Delta/2E_k$ and $n_k = \frac{1}{2}(1-\xi_k/E_k)$ with the quasi-particle energy $E_k = \sqrt{\xi_k^2+\Delta^2}$, and the gap equation gives

$$\Delta_{\mathrm{eq}}(\lambda) = \frac{E_{\mathrm{sh}}}{\sinh(1/\lambda)} \;\approx\; 2E_{\mathrm{sh}}\,e^{-1/\lambda} \quad (\lambda \ll 1).$$

The model keeps what matters for the question — a gap that depends exponentially on a coupling that grows exponentially — and omits the dependence of the gap on the momentum; E_{sh} stands for the range of energies over which the attraction acts. The program starts the model in equilibrium at $\lambda = 0.15$, lets the coupling grow as $\lambda = 0.15\,e^{\tau_d}$ up to $\lambda = 0.5$, and integrates Eqs. (7.2) for 2000 energies (check `def-Q2`: doubling the number of energies and halving the step changes neither the smallest ratio nor the final number of broken pairs in the digits given).

Besides $|\Delta|$ the program records the density n_{ex} of **broken pairs**: the number of pairs of states, per unit of the density of states times E_{sh}, that are not in the ground state belonging to the order parameter of the moment. In equilibrium it is zero.

7.3 Results

gsum/E_{sh}	Ξ_d at the start	smallest $\lvert\Delta\rvert/\Delta_{\mathrm{eq}}$	τ_d of the catch-up	Ξ_d there	final $\lvert\Delta\rvert/\Delta_{\mathrm{eq}}$	n_{ex} at the end
1×10^{-4}	0.262	0.958	0.020	0.23	1.0000	6.0×10^{-6}
3×10^{-4}	0.786	0.890	0.069	0.47	0.9998	5.1×10^{-5}
1×10^{-3}	2.62	0.737	0.234	0.52	0.9980	4.4×10^{-4}
3×10^{-3}	7.86	0.539	0.454	0.44	0.9886	2.4×10^{-3}
0.01	26.2	0.328	0.780	0.32	0.9347	0.012
0.03	78.6	0.189	1.172	0.24	0.8076	0.041
0.1	262	0.098	—	—	0.1189	0.022

1. **The faster the deflation, the further the order parameter falls behind** (check `def-Q3`). For a rate of $10^{-4}E_{\mathrm{sh}}$ the order parameter drops to 0.958 of the equilibrium gap of the moment before it recovers; for $10^{-1}E_{\mathrm{sh}}$ it drops to 0.098.
2. **The state catches up when Ξ_d has fallen to a value between 0.23 and 0.52** (check `def-Q4`): in six of the seven runs the order parameter returns to within 10 per cent of the equilibrium gap at such a value. In the fastest run it has not caught up when the coupling reaches 0.5. This confirms the estimate (7.1), and fixes the number that "small" means.
3. **The lag leaves broken pairs behind** (check `def-Q5`). Their density at the end grows with the rate, from 6.0×10^{-6} to 4.1×10^{-2} in the six slower runs. They are excitations of the superfluid that the deflation itself has made; in a system without dissipation they remain.

Reading for a slow deflation. Because Δ_{eq} grows as $e^{-1/\lambda}$ while λ grows only as e^{τ_d}, the parameter Ξ_d falls steeply with time. For any rate there is a moment at which Ξ_d passes through the window of item 2; before it the paired state is out of equilibrium, with an order parameter below the equilibrium

value, and after it the adiabatic history of Sec. 3 applies. The slower the deflation, the earlier that moment and the smaller the gap at which it happens: it is the moment at which the equilibrium gap has grown to between two and four times gsum/λ.

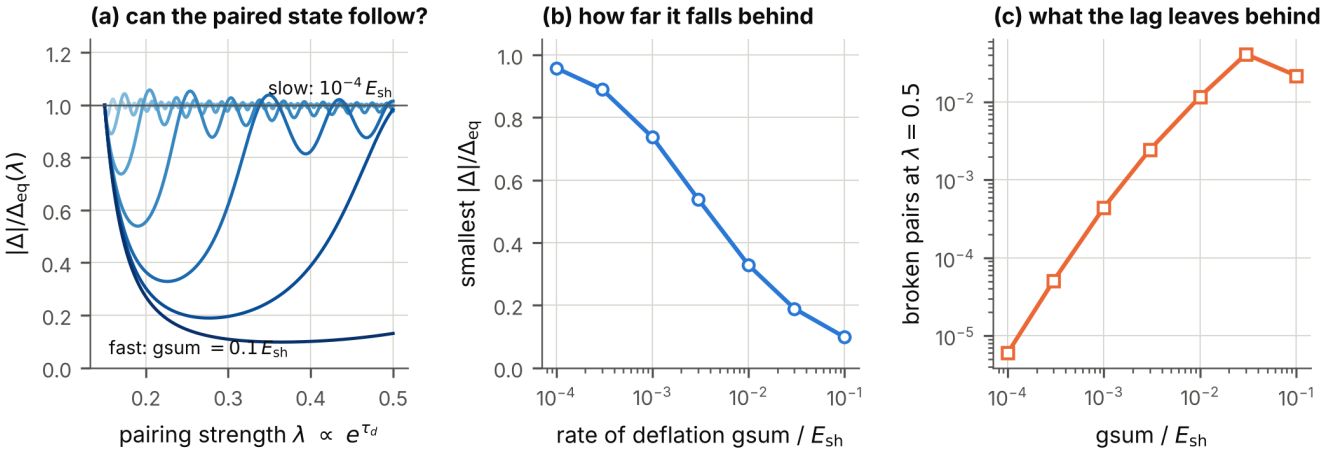


Figure 6: The paired state while the coupling grows. (a) The order parameter divided by the equilibrium gap of the moment, against the pairing strength, for seven rates of deflation. (b) The smallest value of that ratio against the rate. (c) The density of broken pairs at the end of the run against the rate. Data: `data/quench_history.csv`, `data/quench_summary.csv`; program: `deflation quench`.

---

8. What has been established, and what has not

Established.

1. For fields that do not depend on the extra coordinates, the model of the main project on a deflating spacetime is the same model with the couplings (1.2) of the moment, exactly (Sec. 1). The masses are constant; the coupling g_{VS} grows as $e^{\tau_d/2}$; α_{VS}, G_{VS}, D_{VS} and C grow as e^{τ_d}.
2. The modes of the Vorton-Scalar: growth with the weight only, a static potential of unchanged range and of the strength of the moment, thermal creation on 7+1 and extinction on 4+4 for modes that depend on a deflating coordinate (Sec. 2).
3. The exchange amplitudes of the 30 pair channels computed are those of the main project times the square of the coupling of the moment; the most attractive channel is unchanged (Sec. 3.1).
4. The gap grows according to $d\ln(\text{gap})/d\tau_d = 1/\lambda$ in weak coupling; the critical temperature follows it (Sec. 3.2).
5. Two fermions become bound at a definite time, at a scattering resonance, with a binding energy that starts quadratically; the order of the other events depends on m_s/m alone (Sec. 4).
6. The potential of the order parameter prefers no direction among the four frame axes; the ground states are the unitary states (Sec. 5).
7. A vortex line of unit winding carries four fermion zero modes, two for each direction along the line, on both spacetimes and at every moment; its energy per length falls as $e^{-\tau_d}$ (Sec. 6).
8. The paired state follows the deflation if and only if $\Xi_d = \text{gsum}/(\lambda\Delta_{\text{eq}})$ is below about 0.2 to 0.5; before that it lags and accumulates broken pairs (Sec. 7).

Not established.

1. *The end of the history.* The couplings grow without bound. The four-dimensional description with one-boson exchange holds only while α_{VS} is not large, and the eight-dimensional theory itself is an effective theory (its couplings have negative dimensions of mass, S1, Sec. 1.3) that must fail when the proper size of the deflating coordinates reaches its scale of validity. What happens then is outside the model.

2. *The source and the energy.* The deflation is prescribed (DMD, Part 6). The energy of matter is not conserved while the couplings change; where it goes — into the agent that drives the deflation — is not described.
3. *Dissipation.* The model of Sec. 7 has none. With dissipation the broken pairs would relax; without a calculation of the relaxation time nothing can be said about how long they survive.
4. *The dynamics at the binding.* Sec. 4.1 shows that the pair cannot follow the deflation adiabatically at the moment of binding. The corresponding time-dependent two-body problem — how much of the population ends in the bound state — has not been solved.
5. *Vortons.* The ingredients are established (Sec. 6); the equilibrium of a loop on the deflating spacetime, and its stability, are not.
6. *Quantities beyond the mean field.* As in the main project, the gap equation is the two-loop truncation of the effective action (supplement RS4), valid in weak coupling. The late part of every history, where the gap is comparable with the Fermi energy, is outside its range of validity; the numbers there indicate the trend only.
7. *The choice of the fields.* That the fields do not depend on $x0$ is assumed, as in the main project. That they do not depend on $x5, x6, x7$ is derived on 4+4 (up to the condition of admissibility discussed in DMD, Part 6) and assumed on 7+1, where modes that depend on those coordinates are populated thermally at the temperature T_{defl}, which is negligible for any allowed rate (RS3).

Appendix A. Complete output of the programs

A.1 `deflation history`

```
PROJECT spacetimes, Revision_Deflation: checks of the history of the model along the deflation
 ↪  clock (couplings that grow as exp(gsum x4))
generated by: deflation (Rust crate spacetimes-deflation, version 0.1.0; PROJECT spacetimes,
 ↪  Revision_Deflation); command: deflation history
8 of 8 checks passed

[PASS] def-H1  (numerical) thresholds of the model, computed again: a pair of fermions is bound if
 ↪  D_VS > 1.67981; matter is self-bound if C > 3.7011 (3 pi^2/8 = 3.7011) or, with the
 ↪  negative-energy states, C > 5.9218 (3 pi^2/5 = 5.9218). On a deflating spacetime D_VS and C
 ↪  grow as exp(gsum x4), so each threshold is passed at a definite x4
[PASS] def-H2  (numerical) along the clock, wherever the gap is below 0.05 E_F the critical
 ↪  temperature is 0.567 times the gap (largest deviation 0.0020); the gap of the vector channel
 ↪  is at every moment at least as large as the gap with the radial mode alone, and that at least
 ↪  as large as the gap of the two-form channel; the gap never decreases as the deflation proceeds
 ↪  (four sets of m_s/m and k_F/m, 34 moments each)
[PASS] def-H3  (numerical) growth of the gap: for a weakly coupled state the gap is proportional
 ↪  to exp(-1/lambda), where lambda = lambda_S + lambda_P is the sum of the pairing strengths of
 ↪  the radial mode and of the phase mode in the vector channel, and lambda grows as exp(tau_d),
 ↪  which gives d ln(gap)/d tau_d = 1/lambda; in the computed histories lambda d ln(gap)/d tau_d
 ↪  lies between 1.00 and 1.03 wherever 1e-6 < gap/E_F < 0.02: the gap grows much faster than the
 ↪  couplings, by one e-folding in the time 1/(gsum/lambda)
[PASS] def-H4  (numerical) the moment at which a pair becomes bound: just below D_c the scattering
 ↪  length is large and negative (-134.0/m_s at 0.99 D_c), just above large and positive
 ↪  (135.9/m_s at 1.01 D_c); the binding energy starts as a power of the distance from the
 ↪  threshold with the exponents 2.01 (between 2 and 4 per cent above D_c) and 2.01 (between 4 and
 ↪  8 per cent), that is quadratically: E_b ~ (exp(tau_pair) - 1)^2 m_s^2/m. Because E_b vanishes
 ↪  at the threshold, the pair cannot follow the deflation adiabatically at first, however slow
 ↪  the deflation is
```

```
[PASS] def-H5  (numerical) matter of fixed density loses the fermion mass (the Vorton-Scalar loses
 ↪  its expectation value in the matter) when C passes a critical value that falls with the
 ↪  density: k_F = 0.2 m: C = 151.892 (2.46 times pi^2 m^2/(4 k_F^2)); k_F = 0.5 m: C = 12.971
 ↪  (1.31 times pi^2 m^2/(4 k_F^2)); k_F = 1 m: C = 2.556 (1.04 times pi^2 m^2/(4 k_F^2)); k_F = 2
 ↪  m: C = 0.617 (1.00 times pi^2 m^2/(4 k_F^2)). The estimate C = pi^2 m^2/(4 k_F^2) of the main
 ↪  project is reached at high density, where the transition happens at weak coupling; at low
 ↪  density the critical C is larger. On a deflating spacetime every piece of matter becomes
 ↪  abnormal at a definite x4, the denser the earlier
[PASS] def-H6  (numerical) the field beyond which the one-loop potential of the Vorton-Scalar lies
 ↪  below its vacuum value decreases monotonically as C grows: 24.7 v_VS at C = 1, 1.77 v_VS at C
 ↪  = 10. As the deflation proceeds the barrier that protects the vacuum moves in towards the
 ↪  vacuum value
[PASS] def-H7  (numerical) the temperature T_half at which the fermions lose half of their mass,
 ↪  in units of the (constant) fermion mass, falls monotonically as g_VS grows, for m_s/m = 0.3, 1
 ↪  and 3 (26 of the 30 moments examined; it is of the order of the four-dimensional vacuum value
 ↪  m/g_VS, which falls as exp(-gsum x4/2)): matter of a fixed temperature melts at a definite x4.
 ↪  No melting temperature can be given for m_s/m = 0.3 from g_VS = 1.69 (C = 32) on; there the
 ↪  Hartree potential ceases to exist at 1.05 v_VS before the expectation value has fallen,
 ↪  because the one-loop potential turns over too close to the vacuum value
[PASS] def-H8  (numerical) order of events: the pair of fermions becomes bound before the one-loop
 ↪  potential of the Vorton-Scalar has fallen below its vacuum value at a field of 3 v_VS only for
 ↪  the mass ratios m_s/m = 10, 30 of the six ratios 0.1, 0.3, 1, 3, 10, 30 examined (the
 ↪  condition is 4 pi D_c m/m_s < 4.27, i.e. m_s/m > 4.9): for a light radial mode the deflation
 ↪  destabilizes the vacuum first
[PASS] ALL  (numerical) every check above passed
```

A.2 `deflation channels`

```
PROJECT spacetimes, Revision_Deflation: checks of the exchange of the Vorton-Scalar on the
 ↪  deflating spacetimes (def-K..: amplitudes in the pair channels; def-B..: the exchanged field)
generated by: deflation (Rust crate spacetimes-deflation, version 0.1.0; PROJECT spacetimes,
 ↪  Revision_Deflation); command: deflation channels
7 of 7 checks passed

[PASS] def-K1  (numerical) the exchange amplitudes of the Vorton-Scalar in the 30 representative
 ↪  pair channels of the main project (15 on each spacetime; three momenta and 13 scattering
 ↪  angles each: 1170 rows) computed again with the frame matrices agree with every number of the
 ↪  file data/pair_channel_kernel.csv of the main project to the digits printed there (1170 rows
 ↪  compared). For fields that do not depend on x5, x6, x7 the amplitude of a channel on the
 ↪  deflating spacetime is the flat amplitude times the square of the coupling of the moment
[PASS] def-K2  (numerical) the amplitudes of the hidden-axis channel reproduce the closed forms of
 ↪  the main project (largest deviation 1.1e-15)
[PASS] def-K3  (numerical) the quartic invariant of the gap matrix of the four vector pairs,
 ↪  Tr[(M_gap M_gap^dagger)^2]/8 = 2 (sum |Delta^a|^2)^2 - |eta_ab Delta^a Delta^b|^2 with the
 ↪  frame metric of x0, x5, x6, x7, holds on both deflating spacetimes (largest deviation
 ↪  2.1e-14); in coordinate components Delta_t = b_t(x4) Delta^t the same form reads with the
 ↪  metric g_tt of the moment
[PASS] def-B1  (numerical) the field that a static source builds up from zero approaches exp(gsum
 ↪  x4)/(k^2 + m_bos^2): after tau_d = gsum x4 = 20 the relative difference is 1.9e-5 4.3e-6
 ↪  2.4e-5 3.5e-5 for (k^2 + m_bos^2, gsum) = (1, 0.2), (0.25, 0.2), (1, 1), (4, 0.5) (transient
 ↪  proportional to exp(-tau_d/2)). The potential between two fermions at rest is therefore the
 ↪  Yukawa potential of the unshifted mass m_s with the strength alpha_VS of the moment
[PASS] def-B2  (exact) a mode of the Vorton-Scalar that does not depend on x5, x6, x7 goes as
 ↪  exp(exponent x4) with exponent = gsum/2 +- sqrt(gsum^2/4 - k^2 - m_bos^2): it oscillates if
 ↪  and only if k^2 + m_bos^2 > gsum^2/4, and its amplitude then grows as exp(gsum x4/2) (the
 ↪  weighted field has constant amplitude: the number of quanta is conserved). The phase mode
 ↪  (m_bos = 0) does not oscillate for k < gsum/2; the radial mode oscillates at every k if m_s >
 ↪  gsum/2 (table data/scalar_zero_modes.csv)
```

```
[PASS] def-B3  (numerical) 7+1: a mode of the Vorton-Scalar that depends on one deflating
 ↪  coordinate and starts empty ends with 1/(exp(2 pi varpi_b/rate_t) - 1) quanta, a Bose-Einstein
 ↪  law with the temperature rate_t/(2 pi): 36 cases (varpi_b/rate_t = 0.25 to 3, each of the
 ↪  three coordinates; the number is read off where the physical wavenumber is 3000 varpi_b),
 ↪  largest relative deviation 1.9e-5; the Bogoliubov coefficients satisfy |alpha|^2 - |beta|^2 =
 ↪  1 to 1.3e-13
[PASS] def-B4  (numerical) 4+4: a mode of the Vorton-Scalar that depends on a deflating time has
 ↪  one bounded solution; at early x4 it is a standing wave (the two travelling waves have equal
 ↪  amplitudes) and its profile is the modified Bessel function K of the imaginary order i
 ↪  varpi_b/rate_t (ratio to the exact form constant to 1e-4 away from the zeros): varpi_b/rate_t
 ↪  = 0.5: |c_-/c_+| - 1 = 0.0e0, 23 of 23 samples; varpi_b/rate_t = 1.5: |c_-/c_+| - 1 = 0.0e0,
 ↪  24 of 24 samples; varpi_b/rate_t = 3: |c_-/c_+| - 1 = 0.0e0, 26 of 26 samples;
[PASS] ALL  (numerical) every check above passed
```

A.3 `deflation quench`

```
PROJECT spacetimes, Revision_Deflation: checks of the time-dependent dynamics of the paired state
 ↪  under a coupling that grows as exp(gsum x4)
generated by: deflation (Rust crate spacetimes-deflation, version 0.1.0; PROJECT spacetimes,
 ↪  Revision_Deflation); command: deflation quench
5 of 5 checks passed

[PASS] def-Q1  (numerical) the equations of motion conserve |2F|^2 + (1 - 2n)^2 = 1 for every
 ↪  pair: largest deviation at the end of the seven runs 4.2e-11
[PASS] def-Q2  (numerical) convergence: with twice as many energies (4000) and half the time step
 ↪  the run with gsum = 1e-3 E_sh gives the smallest ratio 0.73710 (against 0.73710) and the final
 ↪  density of broken pairs 4.4255e-4 (against 4.4255e-4)
[PASS] def-Q3  (numerical) the faster the deflation, the further the order parameter falls behind:
 ↪  the smallest value of |Delta|/Delta_eq is 0.958, 0.890, 0.737, 0.539, 0.328, 0.189, 0.098 for
 ↪  gsum/E_sh = 1e-4, 3e-4, 1e-3, 3e-3, 1e-2, 3e-2, 1e-1 (Xi_d at the start: 0.26 to 262)
[PASS] def-Q4  (numerical) the order parameter catches up with the equilibrium gap of the moment
 ↪  (to within 10 per cent) when Xi_d = gsum/(lambda Delta_eq) has fallen to a value between 0.23
 ↪  and 0.52 (six runs; in the fastest run, gsum = 0.1 E_sh, it does not catch up before lambda =
 ↪  0.5): the paired state follows the deflation if and only if gsum is small compared with lambda
 ↪  times the gap
[PASS] def-Q5  (numerical) the lag leaves broken pairs behind: their density at lambda = 0.5 is
 ↪  6.0e-6, 5.1e-5, 4.4e-4, 2.4e-3, 1.2e-2, 4.1e-2 (units of E_sh times the density of states) for
 ↪  gsum/E_sh = 1e-4 to 3e-2, growing with the rate; for the equilibrium state it is zero
[PASS] ALL  (numerical) every check above passed
```

A.4 `deflation symmetry`

The checks `44d-S..`, `71d-S..` are discussed in DMD, Part 5, and in the supplement RS2.

```
PROJECT spacetimes, Revision_Deflation: checks of the symmetries on the deflating spacetimes
 ↪  (isometries, internal symmetry of the fields that do not depend on the extra coordinates, zero
 ↪  modes on a vortex line, reduction of the couplings)
generated by: deflation (Rust crate spacetimes-deflation, version 0.1.0; PROJECT spacetimes,
 ↪  Revision_Deflation); command: deflation symmetry
18 of 18 checks passed

[PASS] 44d-S1  (exact) of the 28 rotations and boosts of the flat 4+4 spacetime (one for each pair
 ↪  of coordinates) exactly 6 remain isometries of the deflating spacetime for three different
 ↪  rates (the rotations of x0, x1, x2, x3), 7 if two rates are equal (in addition the rotation of
 ↪  the two coordinates that deflate at the same rate) and 9 if the three rates are equal (in
 ↪  addition the three rotations of x5, x6, x7); no transformation that involves x4 survives, and
 ↪  none that mixes x0 with a deflating coordinate. Together with the 7 translations, the scaling
 ↪  and, for equal rates, the three special vectors of metric.rs this gives 14, 15 and 20 Killing
 ↪  vectors (Lie derivative of the metric zero at two points, exact arithmetic)
```

```
[PASS] 44d-S2  (exact) the products gamma^A gamma^B that commute with the five matrices alpha^1,
↪  alpha^2, alpha^3, beta, i beta chi (and so act as internal symmetries on fields that do not
↪  depend on the extra coordinates) are exactly those of the six pairs taken from x0, x5, x6, x7.
↪  Of these six, the deflating spacetime keeps as isometries 0 (three different rates), 1 (two
↪  equal rates) and 3 (three equal rates): the remaining 6, 5 and 3 are symmetries of the sector
↪  of such fields that have no counterpart among the isometries of the spacetime (hidden
↪  symmetries)
[PASS] 71d-S1  (exact) of the 28 rotations and boosts of the flat 7+1 spacetime (one for each pair
↪  of coordinates) exactly 6 remain isometries of the deflating spacetime for three different
↪  rates (the rotations of x0, x1, x2, x3), 7 if two rates are equal (in addition the rotation of
↪  the two coordinates that deflate at the same rate) and 9 if the three rates are equal (in
↪  addition the three rotations of x5, x6, x7); no transformation that involves x4 survives, and
↪  none that mixes x0 with a deflating coordinate. Together with the 7 translations, the scaling
↪  and, for equal rates, the three special vectors of metric.rs this gives 14, 15 and 20 Killing
↪  vectors (Lie derivative of the metric zero at two points, exact arithmetic)
[PASS] 71d-S2  (exact) the products gamma^A gamma^B that commute with the five matrices alpha^1,
↪  alpha^2, alpha^3, beta, i beta chi (and so act as internal symmetries on fields that do not
↪  depend on the extra coordinates) are exactly those of the six pairs taken from x0, x5, x6, x7.
↪  Of these six, the deflating spacetime keeps as isometries 0 (three different rates), 1 (two
↪  equal rates) and 3 (three equal rates): the remaining 6, 5 and 3 are symmetries of the sector
↪  of such fields that have no counterpart among the isometries of the spacetime (hidden
↪  symmetries)
[PASS] def-S3  (numerical) the internal symmetry of the fields that do not depend on the extra
↪  coordinates, computed again for both spacetimes (24 items of data/reduced_symmetry.csv, all
↪  as expected): the commutant of the five matrices has dimension 8 and is u(2) + u(2) = su(2)_A
↪  + su(2)_B + u(1)_F + u(1)_E; the eight one-particle states are (2,1) + (1,2); the four vector
↪  pair operators are the (2,2) with fermion number 2, and their invariant quadratic form is the
↪  frame metric of the axes x0, x5, x6, x7. The deflation does not change this symmetry, because
↪  the weighted field of this sector obeys the flat equations (checks 44d-D.., 71d-D.. of the
↪  command matrices)
[PASS] 44d-V1  (exact) alpha^1, alpha^2, alpha^3, beta and i beta chi are Hermitian, have square
↪  one and anticommute with one another
[PASS] 44d-V2  (exact) i beta alpha^1 and alpha^2 beta chi are commuting Hermitian matrices with
↪  square one
[PASS] 44d-V3  (exact) each of the four joint eigenspaces of the two matrices has dimension 4: a
↪  vortex line of unit winding carries four zero modes of the sixteen-component field
[PASS] 44d-V4  (exact) Pc = i alpha^1 alpha^2 alpha^3 chi equals the product of the five matrices,
↪  commutes with each of them, is Hermitian and traceless and has square one: it is, up to sign,
↪  the generator P of U(1)_E
[PASS] 44d-V5  (exact) winding +1: every w_z with (i beta alpha^1) w_z = -w_z, (alpha^2 beta chi)
↪  w_z = -w_z solves the zero-energy equations (i alpha^1 + beta) w_z = 0, (i alpha^2 + i beta
↪  chi) w_z = 0; on these four spinors alpha^3 = Pc, and its trace vanishes: two zero modes have
↪  alpha^3 = +1 (energy +k_3) and two alpha^3 = -1 (energy -k_3)
[PASS] 44d-V6  (exact) winding -1: the spinors w_z with (i beta alpha^1) w_z = -w_z, (alpha^2 beta
↪  chi) w_z = +w_z solve (i alpha^1 + beta) w_z = 0, (i alpha^2 - i beta chi) w_z = 0, and on
↪  them alpha^3 = -Pc: the directions of motion are exchanged
[PASS] 71d-V1  (exact) alpha^1, alpha^2, alpha^3, beta and i beta chi are Hermitian, have square
↪  one and anticommute with one another
[PASS] 71d-V2  (exact) i beta alpha^1 and alpha^2 beta chi are commuting Hermitian matrices with
↪  square one
[PASS] 71d-V3  (exact) each of the four joint eigenspaces of the two matrices has dimension 4: a
↪  vortex line of unit winding carries four zero modes of the sixteen-component field
[PASS] 71d-V4  (exact) Pc = i alpha^1 alpha^2 alpha^3 chi equals the product of the five matrices,
↪  commutes with each of them, is Hermitian and traceless and has square one: it is, up to sign,
↪  the generator P of U(1)_E
[PASS] 71d-V5  (exact) winding +1: every w_z with (i beta alpha^1) w_z = -w_z, (alpha^2 beta chi)
↪  w_z = -w_z solves the zero-energy equations (i alpha^1 + beta) w_z = 0, (i alpha^2 + i beta
↪  chi) w_z = 0; on these four spinors alpha^3 = Pc, and its trace vanishes: two zero modes have
↪  alpha^3 = +1 (energy +k_3) and two alpha^3 = -1 (energy -k_3)
[PASS] 71d-V6  (exact) winding -1: the spinors w_z with (i beta alpha^1) w_z = -w_z, (alpha^2 beta
↪  chi) w_z = +w_z solve (i alpha^1 + beta) w_z = 0, (i alpha^2 - i beta chi) w_z = 0, and on
↪  them alpha^3 = -Pc: the directions of motion are exchanged
```

```
[PASS] def-S4  (exact) reduction to fields that do not depend on x5, x6, x7: sqrt|g| times the
 ↪  Yukawa term and the potential of the Vorton-Scalar, written with the weighted fields psihat =
 ↪  sqrt(Vd) psi and Phihat = sqrt(Vd) Phi_VS, are the same expressions with the couplings
 ↪  y_VS/sqrt(Vd), lambda_VS/Vd and the vacuum value v_VS sqrt(Vd); the products y_VS v_VS and
 ↪  lambda_VS v_VS^2, hence the fermion mass and the mass of the radial mode, do not change
 ↪  (rational sample values, exact). The exponents of data/coupling_exponents.csv follow, with Vd
 ↪  proportional to exp(-gsum x4)
[PASS] ALL  (exact) every check above passed
```

Appendix B. Files

file	content	written by
`code/deflation-rs/src/history.rs`	couplings along the clock; gap, pair, matter, vacuum and temperature at each moment	—
`code/deflation-rs/src/scalar.rs`	modes of the scalar: exponents, static source, creation, bounded solutions	—
`code/deflation-rs/src/quench.rs`	time-dependent mean-field dynamics of the shell model	—
`code/deflation-rs/src/symmetry_defl.rs`	algebra of the vortex zero modes; reduction of the couplings	—
`code/deflation-rs/src/cmd_channels.rs`	exchange amplitudes on the deflating spacetimes; quartic invariant	—
`data/coupling_exponents.csv`	the exponents of Sec. 1.3	`deflation symmetry`
`data/events.csv`	the order of events of Sec. 4.2	`deflation history`
`data/gap_history.csv`	the gap along the clock, four sets	`deflation history`
`data/dimer_history.csv`	scattering length and binding energy along the clock	`deflation history`
`data/pair_channel_kernel.csv`, `data/gl_quartic_check.csv`	exchange amplitudes; the identity (5.1)	`deflation channels`
`data/scalar_zero_modes.csv`, `data/static_source.csv`	Sec. 2.1 and 2.2	`deflation channels`
`data/boson_creation.csv`, `data/boson_bounded.csv`	Sec. 2.3	`deflation channels`
`data/quench_history.csv`, `data/quench_summary.csv`	Sec. 7	`deflation quench`
`data/order_parameter_states.csv`	Sec. 5.2	`deflation paired`
`data/history_checks.txt`, `channel_checks.txt`, `quench_checks.txt`, `symmetry_checks.txt` (and `.csv`)	Appendix A	the four commands
`figures/fig_clock`, `fig_scalar`, `fig_kernels`, `fig_gap_history`, `fig_binding`, `fig_quench` (`.pdf`, `.png`)	the six figures	`make_figures.py`